

Packet Tracer - Troubleshoot Connectivity Issues

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	G0/1	172.16.2.1	255.255.255.0	N/A
	S0/0/0	209.165.200.226	255.255.255.252	N/A
R2	G0/0	209.165.201.1	255.255.255.224	N/A
	S0/0/0 (DCE)	209.165.200.225	255.255.255.252	N/A
PC-01	NIC	172.16.1.3	255.255.255.0	172.16.1.1
PC-02	NIC	172.16.1.4	255.255.255.0	172.16.1.1
PC-A	NIC	172.16.2.3	255.255.255.0	172.16.2.1
PC-B	NIC	172.16.2.4	255.255.255.0	172.16.2.1
Web	NIC	209.165.201.2	255.255.255.224	209.165.201.1
DNS1	NIC	209.165.201.3	255.255.255.224	209.165.201.1
DNS2	NIC	209.165.201.4	255.255.255.224	209.165.201.1

Objectives

In this Packet Tracer activity, you will troubleshoot and resolve connectivity issues, if possible. Otherwise, the issues should be clearly documented so they can be escalated.

Background / Scenario

Users are reporting that they cannot access the web server, www.cisco.pka after a recent upgrade that included adding a second DNS server. You must determine the cause and attempt to resolve the issues for the users. Clearly document the issues and any solution(s). You do not have access to the devices in the cloud or the server www.cisco.pka. Escalate the problem if necessary.

Note: Router R1 can only be accessed using SSH with the username **Admin01** and password **cisco12345**. Router R2 is in the ISP cloud and is not accessible by you.

Instructions

Step 1: Determine connectivity issues from PC-01.

- On PC-01, open the command prompt. Enter the command **ipconfig** to verify what IP address and default gateway have been assigned to PC-01. Correct as necessary according to the Addressing Table.

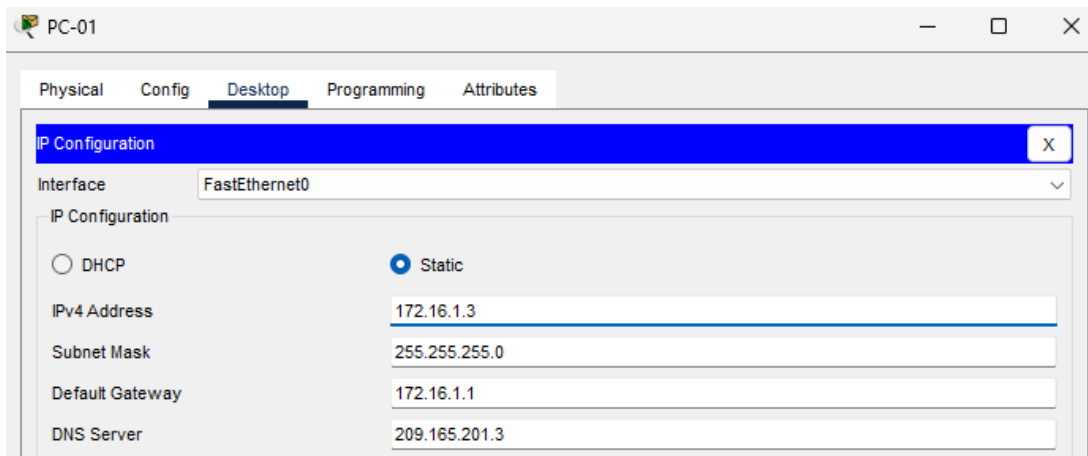
```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address. . . . .: 0030.F2D8.3C0A
    Link-local IPv6 Address . . . . .: FE80::2
    IPv6 Address. . . . .: 2001:DB8:ACAD::2
    IPv4 Address. . . . .: 172.168.1.3
    Subnet Mask. . . . .: 255.255.255.0
    Default Gateway. . . . .: FE80::1
                           172.16.1.1
    DHCP Servers. . . . .: 0.0.0.0
    DHCPv6 IAID. . . . .:
    DHCPv6 Client DUID. . . . .: 00-01-00-01-S2-A7-D4-C9-00-30-F2-D8-3C-0A
    DNS Servers. . . . .: 2001:DB8:CAFE::3
                           209.165.201.3

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address. . . . .: 0060.47B1.DD31
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address. . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask. . . . .: 0.0.0.0
    Default Gateway. . . . .: ::
                           0.0.0.0
    DHCP Servers. . . . .: 0.0.0.0
    DHCPv6 IAID. . . . .:
    DHCPv6 Client DUID. . . . .: 00-01-00-01-S2-A7-D4-C9-00-30-F2-D8-3C-0A
    DNS Servers. . . . .: ::
                           209.165.201.3
```



- b. After verifying/correcting the IP addressing issues on PC-01, issue pings to the default gateway, web server, and other PCs. Were the pings successful? Record the results.

Ping to default gateway (172.16.1.1)?

```
C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time=5ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

Yes

To web server (209.165.201.2)?

```
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Reply from 209.165.201.2: bytes=32 time=9ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 9ms, Average = 3ms
```

Yes

Ping to PC-02?

```
C:\>ping 172.16.1.4

Pinging 172.16.1.4 with 32 bytes of data:

Reply from 172.16.1.4: bytes=32 time<1ms TTL=128
Reply from 172.16.1.4: bytes=32 time<1ms TTL=128
Reply from 172.16.1.4: bytes=32 time<1ms TTL=128
Reply from 172.16.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

To PC-A?

```
C:\>ping 172.16.2.3

Pinging 172.16.2.3 with 32 bytes of data:

Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.

Ping statistics for 172.16.2.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Unreachable

To PC-B?

```
C:\>ping 172.16.2.4

Pinging 172.16.2.4 with 32 bytes of data:

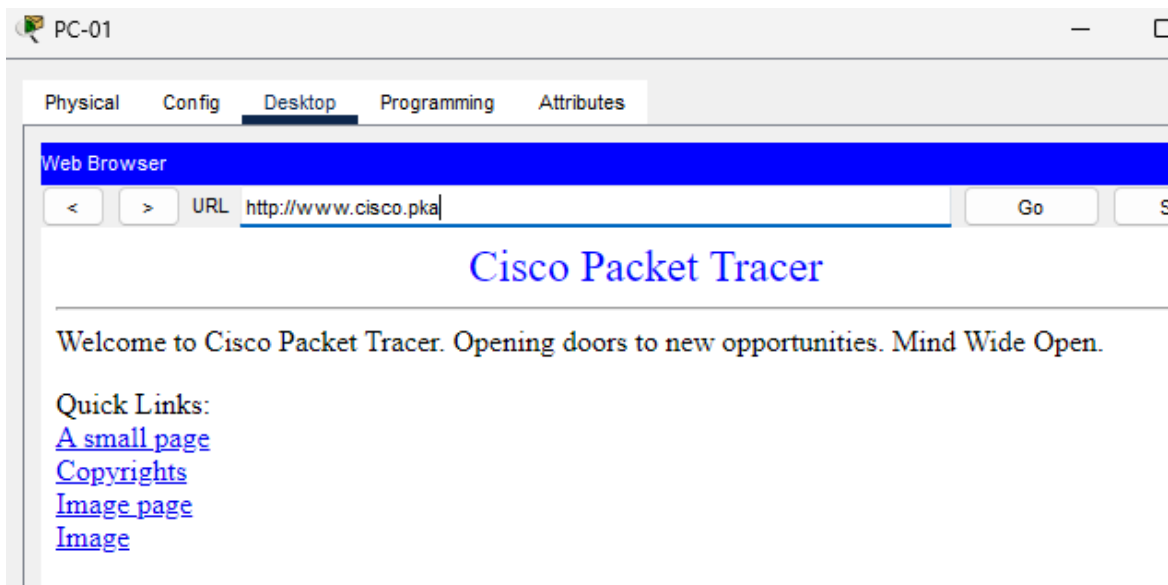
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.

Ping statistics for 172.16.2.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Unreachable

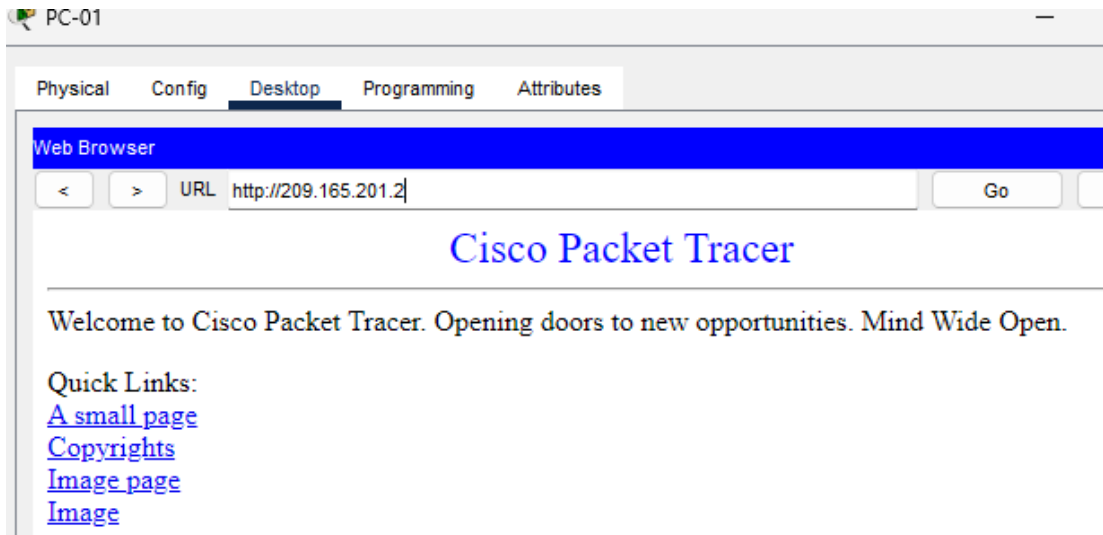
- c. Use the web browser to access the web server on PC-01. Access the web server by first entering the URL <http://www.cisco.pka> and then by using the IP address 209.165.201.2. Record the results.

Can PC-01 access www.cisco.pka?



Yes

Using the web server IP address?



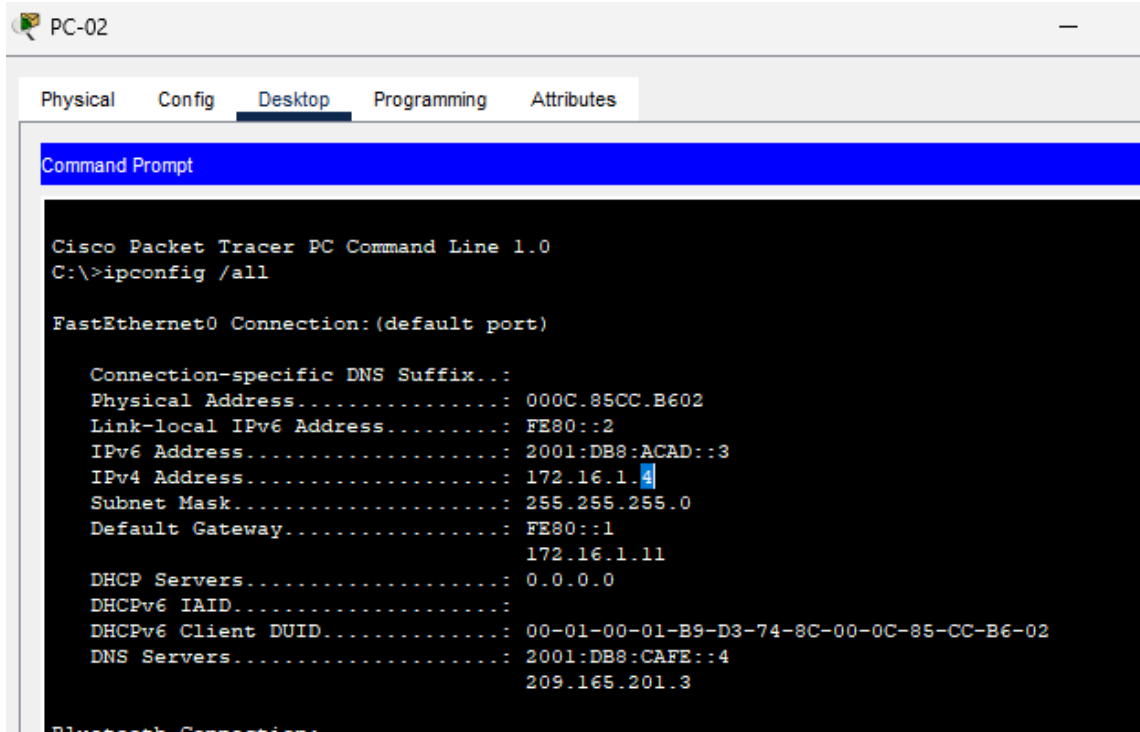
Yes

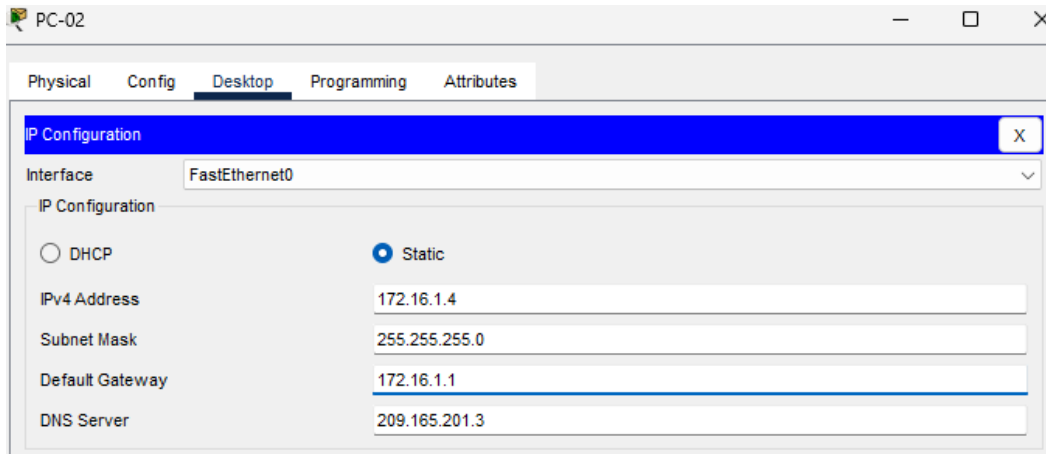
- d. Document the issues and provide the solution(s). Correct the issues if possible.

The IP address on PC-01 is incorrectly configured. To resolve the issue, the IP address is changed from 172.168.1.3 to 172.16.1.3. PC-01 cannot successfully ping the PCs on the 172.16.2.0/24 network.

Step 2: Determine connectivity issues from PC-02.

- a. On PC-02, open the command prompt. Enter the command **ipconfig** to verify the configuration for the IP address and default gateway. Correct as necessary.





- b. After verifying/correcting the IP addressing issues on PC-02, issue pings to the default gateway, web server, and other PCs. Were the pings successful? Record the results.

Ping to default gateway (172.16.1.1)?

```
C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Yes

To web server (209.165.201.2)?

```
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Reply from 209.165.201.2: bytes=32 time=16ms TTL=126
Reply from 209.165.201.2: bytes=32 time=13ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 7ms
```

Yes

Ping to PC-01?

```
C:\>ping 172.16.1.3

Pinging 172.16.1.3 with 32 bytes of data:

Reply from 172.16.1.3: bytes=32 time<1ms TTL=128
Reply from 172.16.1.3: bytes=32 time<1ms TTL=128
Reply from 172.16.1.3: bytes=32 time<1ms TTL=128
Reply from 172.16.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

To PC-A?

```
C:\>ping 172.16.2.3

Pinging 172.16.2.3 with 32 bytes of data:

Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.

Ping statistics for 172.16.2.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Unreachable

To PC-B?

```
C:\>ping 172.16.2.4

Pinging 172.16.2.4 with 32 bytes of data:

Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.
Reply from 209.165.200.225: Destination host unreachable.

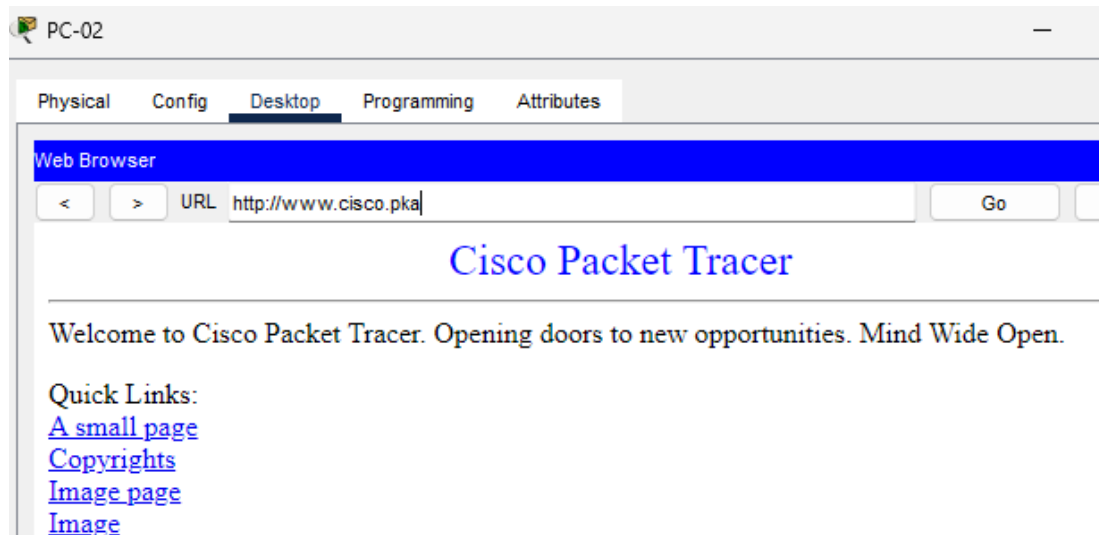
Ping statistics for 172.16.2.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Unreachable

- c. Navigate to www.cisco.pka using the web browser on PC-02. Record the results.

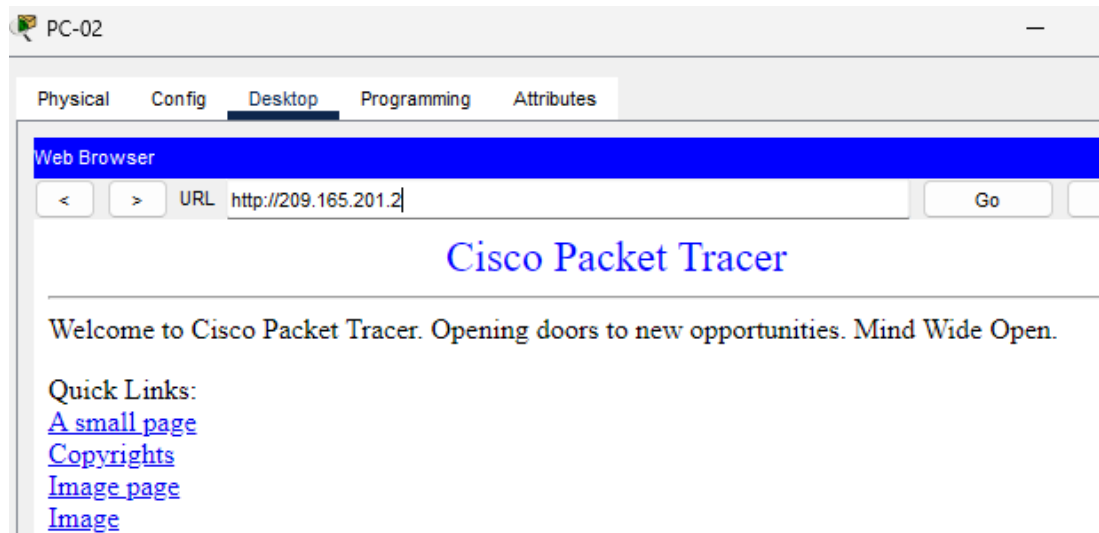
Questions:

Can PC-02 access www.cisco.pka?



Yes

Using the web server IP address?



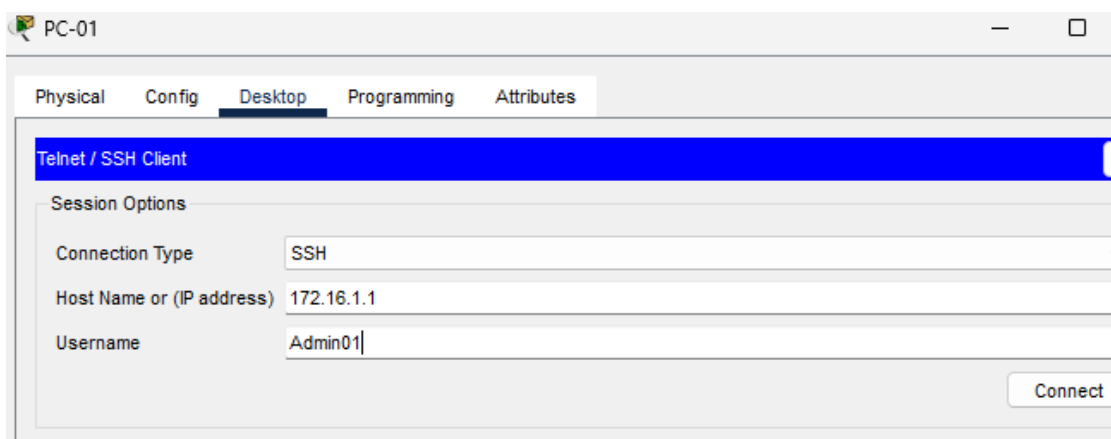
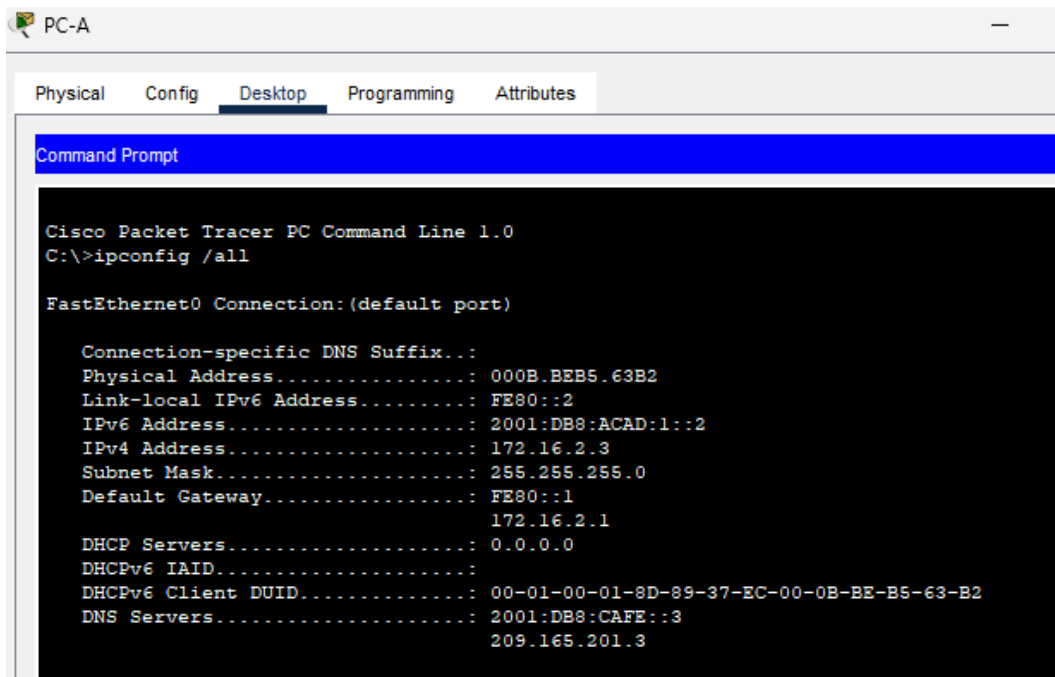
Yes

- d. Document the issues and provide the solution(s). Correct the issues if possible.

PC-02 can access the web server using the IP address after correcting the default gateway. The default gateway should be configured as 172.16.1.1 on PC-02. PC-02 cannot successfully ping the PCs on the 172.16.2.0/24 network.

Step 3: Determine connectivity issues from PC-A.

- On PC-A, open the command prompt. Enter the command **ipconfig** to verify the configuration for the IP address and default gateway. Correct as necessary.



```
Password:

Warning: Unauthorized Access is Prohibited.

R1#en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int g0/1
R1(config-if)#ip add 172.16.2.1 255.255.255.0
R1(config-if)#no sh
R1(config-if)#end
R1#
R1#cop r st
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

- b. After correcting the IP addressing issues on PC-A, issue the pings to the web server, default gateway, and other PCs. Were the pings successful? Record the results.

To web server (209.165.201.2)?

```
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Reply from 209.165.201.2: bytes=32 time=2ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Yes

Ping to default gateway (172.16.2.1)?

```
C:\>ping 172.16.2.1

Pinging 172.16.2.1 with 32 bytes of data:

Reply from 172.16.2.1: bytes=32 time=6ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

Yes

Ping to PC-B?

```
C:\>ping 172.16.2.4

Pinging 172.16.2.4 with 32 bytes of data:

Reply from 172.16.2.4: bytes=32 time<1ms TTL=128
Reply from 172.16.2.4: bytes=32 time<1ms TTL=128
Reply from 172.16.2.4: bytes=32 time<1ms TTL=128
Reply from 172.16.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

To PC-01?

```
C:\>ping 172.16.1.3

Pinging 172.16.1.3 with 32 bytes of data:

Reply from 172.16.1.3: bytes=32 time<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

To PC-02?

```
C:\>ping 172.16.1.4

Pinging 172.16.1.4 with 32 bytes of data:

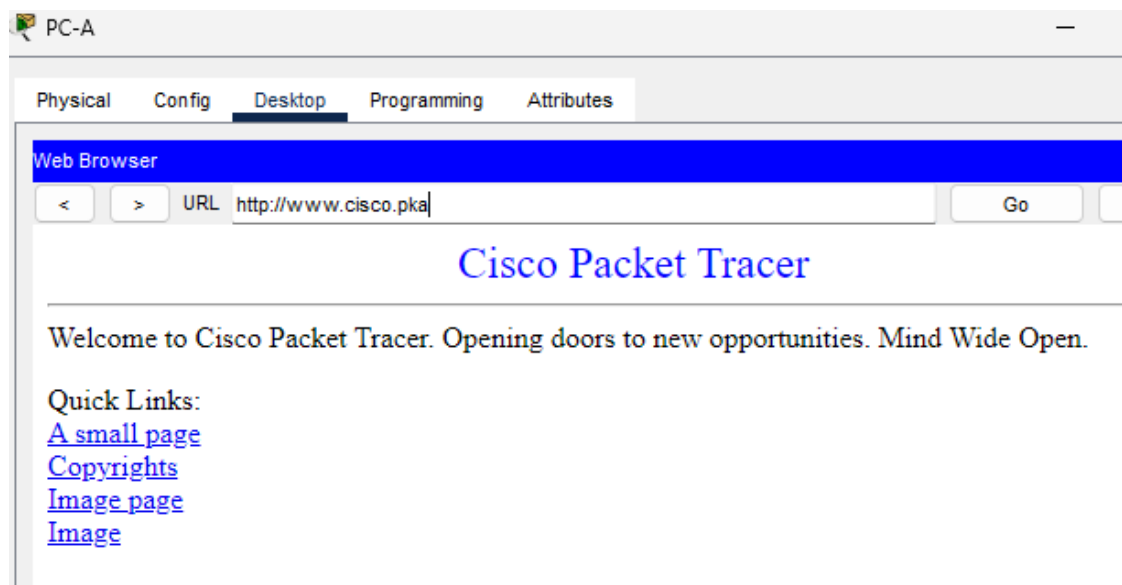
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

- c. Navigate to www.cisco.pka using the web browser on PC-A. Record the results.

Can PC-A access www.cisco.pka?



Yes

Using the web server IP address?



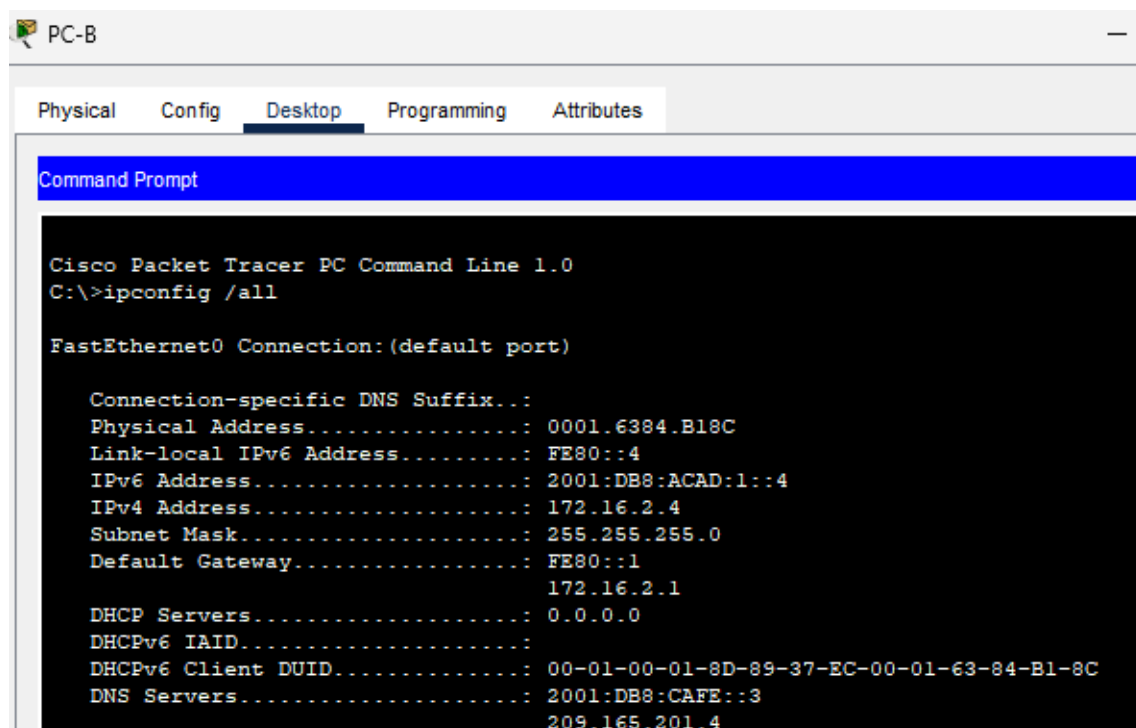
Yes

- d. Document the issues and provide the solution(s). Correct the issues if possible.

PC-A can only access the local LAN. The interface G0/1 on router R1 is incorrectly configured. Correct the IP address on interface G0/1. Access router R1 using SSH from PC-01 or PC-02 to change the IP address from 172.16.3.1 to 172.16.2.1

Step 4: Determine connectivity issues from PC-B.

- a. On PC-B, open the command prompt. Enter the command **ipconfig** to verify the configuration for the IP address and default gateway. Correct as necessary.



- b. After correcting the IP addressing issues on PC-B, issue the pings to the web server, default gateway, and other PCs. Were the pings successful? Record the results.

To web server (209.165.201.2)?

```
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Reply from 209.165.201.2: bytes=32 time=16ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126
Reply from 209.165.201.2: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 4ms
```

Yes

Ping to default gateway (172.16.2.1)?

```
C:\>ping 172.16.2.1

Pinging 172.16.2.1 with 32 bytes of data:

Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

Ping to PC-A?

```
C:\>ping 172.16.2.3

Pinging 172.16.2.3 with 32 bytes of data:

Reply from 172.16.2.3: bytes=32 time<1ms TTL=128
Reply from 172.16.2.3: bytes=32 time<1ms TTL=128
Reply from 172.16.2.3: bytes=32 time=8ms TTL=128
Reply from 172.16.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms
```

Yes

To PC-01?

```
C:\>ping 172.16.1.3

Pinging 172.16.1.3 with 32 bytes of data:

Reply from 172.16.1.3: bytes=32 time<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Yes

To PC-02?

```
C:\>ping 172.16.1.4

Pinging 172.16.1.4 with 32 bytes of data:

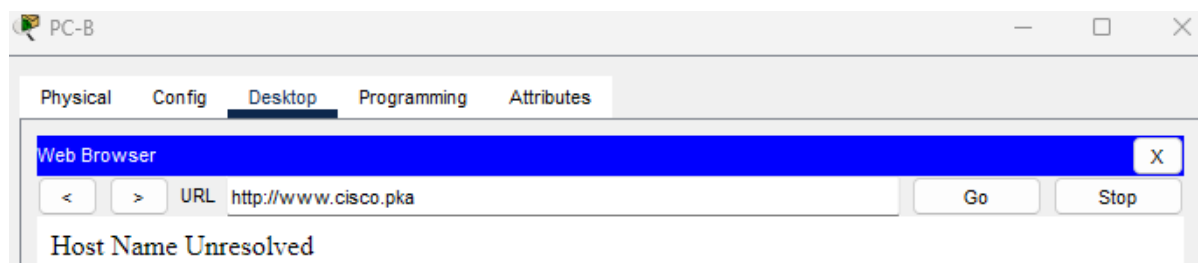
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127
Reply from 172.16.1.4: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Yes

- c. Navigate to www.cisco.pka using the web browser. Record the results.

Can PC-B access www.cisco.pka?



No

Using the web server IP address



Yes

- d. Document the issues and provide the solution(s). Correct the issues if possible.

PC-B can access the web server using the IP address only. PC-B is also configured with the correct DNS-2 server address. This indicates that DNS-2 server may be incorrectly configured. To resolve this issue temporarily, the DNS server address can be configured to use 209.165.200.3. The issue with DNS-2 server needs to be escalated because we do not have administrative access to devices outside your network.

- e. Could all the issues be resolved on PC-B and still make use of DNS2? If not, what would you need to do?

No. DNS2 apparently has configuration issues. You would need to contact the person in charge of the DNS2 server and report your findings.

Step 5: Verify connectivity.

Verify that all the PCs can access the web server www.cisco.pka.

Your completion percentage should be 100%. If not, verify that the IP configuration information is correct on all devices and that it matches what is shown in the addressing table.